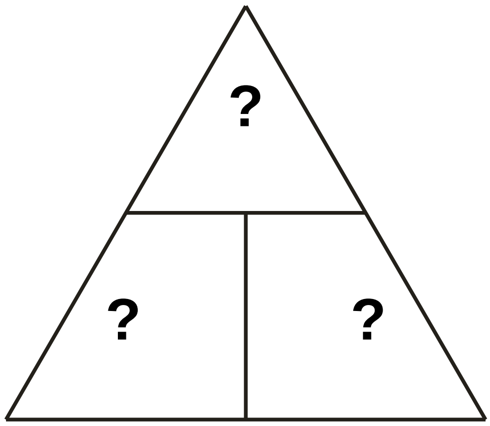
Speed, Distance & Acceleration

Test · 27 marks total

Name: ______________________________ Date: _____________ Score: _____ / 27

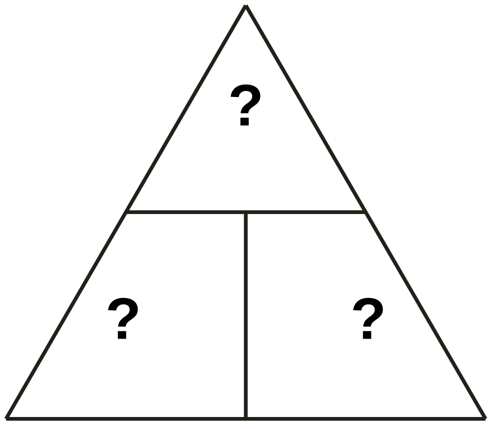
Section A — Complete the formula triangles

A1. Fill in the three blanks with s, v and t in the correct places.



[3 marks]

A2. Fill in the three blanks with Δv, a and t in the correct places.



[3 marks]

Section B — Straight substitution

B1. A cyclist travels at a constant speed of 15 m/s for 8 s. Calculate the distance travelled.

[2 marks]

B2. An object accelerates at 5 m/s^2 for 3 s. Calculate the change in velocity.

[2 marks]

Section C — Rearrange and solve

C1. A car travels 180 m in 12 s. Calculate its average speed. Show how you rearranged the equation.

[3 marks]

C2. A cyclist travels 90 m at a constant speed of 6 m/s. Calculate how long the journey takes. Show how you rearranged the equation.

[3 marks]

C3. An object accelerates from rest to a velocity of 20 m/s in 4 s. Calculate its acceleration.

[3 marks]

Section D — Scalar or vector?

State whether each of the following is a scalar or a vector quantity.

(a) distance (b) displacement (c) speed (d) velocity

[4 marks]

Section E — Graph interpretation

E1. A walker's distance–time graph shows a straight diagonal line for the first 10 s, then a horizontal line for the next 5 s, then a diagonal line steeper than the first for the last 10 s.

(a) Describe the walker's motion during each of the three parts. [3 marks]

(b) State which part shows the fastest speed, and explain how you know. [1 mark]